MACHINE LEARNING – UNITS I TO V

Required Answer Changes, Additions & Numerical Preparation Guide

Purpose: This document consolidates the handwritten instructions and follow-up clarifications visible in the supplied Aditya University question-paper images. It tells you exactly which question needs an addition, what to include, and where to place it in your existing answers.

> **Important: This is an editing and preparation checklist, not a replacement question paper. Keep the original question wording and add the listed material to the corresponding answer.**

Prepared from the supplied Unit I–V question-paper scans and the handwritten notes discussed in this chat.

Contents

UNIT I – INTRODUCTION 2

UNIT II – SUPERVISED LEARNING: BASIC METHODS 2

UNIT III – LINEAR MODELS, REGRESSION AND SUPPORT VECTOR MACHINES 3

UNIT IV – CLASSIFIER EVALUATION AND UNSUPERVISED LEARNING 4

UNIT V – ENSEMBLE LEARNING AND COMPUTATIONAL LEARNING THEORY 6

Final checklist before the exam 7

UNIT I – INTRODUCTION

The handwritten additions below the Unit I question list identify two extra comparison questions.

Q10. Differentiate between Supervised Learning and Unsupervised Learning

Where to change: Add this as a separate answer after Unit I, Q9 (the handwritten Q10).

What to add:

- Use a comparison table only; compare labelled/unlabelled data, objective, output, tasks, evaluation, algorithms and real-world examples.
- Use examples such as house-price prediction or spam classification for supervised learning, and customer grouping for unsupervised learning.

Must include: A clear table with around 8–10 meaningful comparison rows. Do not merge Q10 with Q11.

Q11. Differentiate between Statistical Learning and Machine Learning

Where to change: Add this as a separate answer immediately after Q10 (the handwritten Q11).

What to add:

- Explain the difference in emphasis: statistical learning often emphasizes understanding/interpreting relationships, while machine learning often emphasizes prediction and automated decision-making.
- Mention that the fields overlap and that methods such as Linear Regression and Logistic Regression can appear in both.
- Compare objectives, assumptions, interpretation, model examples, evaluation and applications.

Must include: A comparison table only, followed by a brief accuracy note that the fields are related and not completely separate.

> **Accuracy note: Do not claim that statistical learning never predicts or that machine learning never explains relationships.**

UNIT II – SUPERVISED LEARNING: BASIC METHODS

Q4. Bayesian Learning; MLE vs MAP

Where to change: Expand the existing Bayesian Learning answer with a small worked numerical illustration. The handwritten note beside this area indicates that the topic needs further worked explanation; no exact dataset or values are supplied in the scan.

What to add:

- Explain prior, likelihood, evidence and posterior using Bayes' theorem.
- Show MLE as the parameter value that maximizes likelihood:

$$\theta_{\text{MLE}} = \arg\max_{\theta} P(D \mid \theta).$$

- Show MAP as the parameter value that maximizes posterior:

$$\theta_{\text{MAP}} = \arg\max_{\theta} P(D \mid \theta)P(\theta).$$

- Add one simple numerical example (for example, estimating a coin's probability of heads) and explicitly show how a prior can make MAP differ from MLE.

Must include: Bayes' theorem, MLE and MAP formulas, one worked example, and a comparison table.

> **The scan does not provide a fixed numerical dataset for Q4. Label any numbers used as an illustrative example.**

Q5. k-Nearest Neighbour (kNN), distance metrics and curse of dimensionality

Where to change: Add a dedicated subsection titled "Distance-Weighted kNN – Numerical Example" after the normal kNN algorithm and before the curse of dimensionality.

What to add:

- First explain ordinary kNN: calculate distances, choose K nearest neighbours and use majority voting.
- Explain Euclidean, Manhattan and Minkowski distance formulas.
- For distance-weighted kNN, use a weight such as $w_i = 1/d_i$ (or clearly state another chosen weighting rule).
- Show a numerical example: neighbours at distances 1, 2 and 4 with classes Pass, Fail and Fail. Weights become 1, 0.5 and 0.25. Pass total = 1; Fail total = 0.75; predict Pass.
- Then explain the curse of dimensionality and remedies such as feature selection and PCA.

Must include: Distance calculations/values, weights, class-wise weight totals and the final predicted class.

> **State the weighting formula used. A zero-distance neighbour needs special handling; do not divide by zero.**

Q6. Apply Find-S to positive and negative training examples

Where to change: Keep the existing Find-S procedure, but make the worked application explicit if a dataset is supplied.

What to add:

- Start with the most specific hypothesis (often $\varnothing$ for every attribute).
- Process only positive examples to generalize the hypothesis when needed.
- Show the hypothesis after each positive example; explain that negative examples are not used to update Find-S.
- Write the final most-specific hypothesis consistent with the positive examples.

Must include: A step-by-step hypothesis-update table and the final hypothesis.

> **Find-S ignores negative examples; do not claim it uses them to eliminate hypotheses.**

UNIT III – LINEAR MODELS, REGRESSION AND SUPPORT VECTOR MACHINES

Q7. Logistic Regression: real-world binary classification

Where to change: Add a separate worked numerical subsection after the working procedure. The handwritten note specifically says to practise when values for bias and weight are given.

What to add:

- Write $z = b + wx$ (or $z = b + \sum_i w_i x_i$ for multiple features).
- Apply the sigmoid:
$$P(y = 1) = \frac{1}{1 + e^{-z}}.$$
- Use the sample values $x = 2$, $w = 2$, $b = -3$:
$$z = -3 + (2 \times 2) = 1.$$
- Calculate:
$$P = \frac{1}{1 + e^{-1}} \approx 0.7311 = 73.11\%.$$
- With threshold 0.5, classify as class 1 because $0.7311 \geq 0.5$. State that this is an illustrative calculation.

Must include: Given values → linear score z → sigmoid probability → threshold decision.

> **If the exam supplies different bias/weight/input values, substitute those values rather than memorizing the example result.**

Other Unit III questions

No additional handwritten numerical instruction is clearly visible for the other Unit III questions in the supplied scan. Continue to answer those according to the printed question. For the patient classification and SVM questions, do not invent a prediction for a new patient unless the model, parameters or a valid calculation are provided.

UNIT IV – CLASSIFIER EVALUATION AND UNSUPERVISED LEARNING

Q3. ROC Curve – numerical construction of ROC coordinates

Where to change: After defining ROC, TPR and FPR, add a numerical table that calculates coordinates at multiple thresholds. The handwritten note says "Construct ROC Coordinates".

What to add:

- Use:
$$TPR = \frac{TP}{TP + FN}, \qquad FPR = \frac{FP}{FP + TN}.$$
- For each threshold, calculate TPR and FPR from its confusion-matrix counts.
- Write each ROC coordinate as (FPR, TPR).
- Plot FPR on the X-axis and TPR on the Y-axis; include $(0, 0)$ and $(1, 1)$ endpoints where appropriate.
- Explain AUC and the diagonal random-guessing baseline.

Must include: A threshold-wise calculation table and a correctly labelled ROC plot.

> **The question paper does not provide threshold data. Any table of counts must be labelled as an illustrative example, not as supplied exam data.**

Q6. Confusion matrix – Accuracy, Precision, Recall and F1-score

Where to change: Retain the given values $TP = 40$, $FN = 10$, $FP = 5$, $TN = 45$. The handwritten note indicates that more than one combination may be practised; prepare the formulas so they can be reused with other values.

What to add:

- Show the confusion matrix layout.
- Calculate Accuracy:
$$\frac{TP + TN}{TP + TN + FP + FN} = \frac{40 + 45}{100} = 85\%.$$
- Calculate Precision:
$$\frac{TP}{TP + FP} = \frac{40}{45} = 88.89\%.$$
- Calculate Recall:
$$\frac{TP}{TP + FN} = \frac{40}{50} = 80\%.$$
- Calculate F1:
$$F1 = \frac{2PR}{P + R} \approx 84.21\%.$$
- Practise by substituting a second set of TP, TN, FP and FN values if provided by the teacher.

Must include: Formula, substitution, arithmetic and final percentage for all four metrics.

> **Do not confuse FN with FP. Precision uses FP; Recall uses FN.**

Q10. DBSCAN on geographical customer coordinates

Where to change: Add a full numerical solution using the C1–C8 coordinate table on page 5. The handwritten note specifies $\varepsilon = 1.5$ and MinPts=3, and asks for neighbourhoods, core/border/noise points and number of clusters.

What to add:

- Use Euclidean distance:
$$d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$
- For $\varepsilon = 1.5$, list each point's ε-neighbourhood, counting the point itself.
- Use MinPts=3, counting the point itself, to identify core points.
- Identify border points (non-core points within the ε-neighbourhood of a core point) and noise points.
- State the final clusters and the total number of clusters.

Must include: For the supplied coordinates: Cluster 1 = C1,C2,C3,C4; Cluster 2 = C5,C6,C7; C8 = noise; total clusters = 2. All seven points in the two groups are core points under the stated convention; there are no border points.

> **This result depends on MinPts=3 and on counting the point itself. The printed question gives $\varepsilon = 1.5$; MinPts=3 comes from the handwritten instruction.**

UNIT V – ENSEMBLE LEARNING AND COMPUTATIONAL LEARNING THEORY

Q5. PAC Learnability and VC Dimension

Where to change: Add numerical practice to the theory answer. The handwritten note explicitly lists PAC and VC Dimension as numerical topics.

What to add:

- PAC: identify ε (maximum tolerated error) and δ (failure probability); confidence is $1 - \delta$.
- Use the PAC condition:
$$P(error(h) \leq \varepsilon) \geq 1 - \delta.$$
- Practise a sample-complexity question if a formula and values are supplied. State which PAC bound is being used because different assumptions give different bounds.
- VC Dimension: determine the maximum number of points that a hypothesis class can shatter.
- For a 2D linear classifier, the VC dimension is 3; explain that three non-collinear points can be shattered, but not every labelling of four points.

Must include: PAC parameter substitution and interpretation; VC shattering explanation or diagram.

> **Do not use a sample-complexity formula without stating its assumptions. PAC learnability is not simply the same as VC dimension.**

Q4. Training Error vs Testing Error and Generalization

Where to change: Add a small numerical calculation of training error because the handwritten note lists Training Error as a numerical practice topic.

What to add:

- Training Error = incorrect training predictions / total training samples.
- Testing Error = incorrect test predictions / total test samples.
- Example: 10 incorrect out of 100 training cases gives 10% training error; 20 incorrect out of 100 test cases gives 20% testing error.
- Explain that a large gap between low training error and high testing error may indicate overfitting.
- Explain generalization as performance on unseen data.

Must include: Show numerator, denominator and percentage for both training and testing error.

> **Do not say testing error must always be higher; it often is, but this is not guaranteed.**

Q2 / Q3 / Q6 / Q7 / Q8. Ensemble learning, Random Forest, AdaBoost, Bagging and Boosting

Where to change: The handwritten note explicitly lists Random Forest for numerical practice. Add a short majority-voting calculation to the Random Forest answers; retain the theory already written for Bagging and Boosting.

What to add:

- Show predictions from several Decision Trees for one new input.

- Count votes for each class and select the class with the most votes.
- Example: five trees predict Approved, Rejected, Approved, Approved, Rejected. Votes: Approved=3, Rejected=2; final class=Approved.
- For Bagging, remember independent models trained on bootstrap samples; for Boosting, remember sequential learners that focus on earlier errors.

Must include: A prediction table, vote totals and final majority-vote result.

> **The note does not provide a particular Random Forest dataset or tree outputs. The values above are illustrative.**

Final checklist before the exam

- Unit I: Prepare the two handwritten comparison questions (Q10 and Q11).
- Unit II: Practise Bayesian MLE/MAP worked explanation, distance-weighted kNN with values, and Find-S update steps.
- Unit III: Practise Logistic Regression when bias, weight and input values are given.
- Unit IV: Practise ROC coordinate construction, confusion-matrix calculations and the full DBSCAN coordinate problem.
- Unit V: Practise PAC numerical interpretation, VC shattering, training/testing error calculations and Random Forest majority voting.
- In every numerical answer, show the formula, substitution, calculation and final result.
- Keep illustrative values clearly separate from values actually supplied in the question paper.